Propiedades del sistema de tipos - Son teoremas

Unicidad de Tipos

$$Si\ \Gamma\ \vdash\ \mathrm{M}:\ \sigma\ y\ \Gamma \vdash \mathrm{M}:\beta\ \ son\ derivables,\ entonces\ \sigma\ =\ \beta$$

Weakening + Strengthening

$$Si\ \Gamma\ \vdash\ \mathrm{M}:\ \sigma\ es\ derivable\ y\ \mathrm{fv(M)}\ \subseteq\ \mathrm{dom}(\Gamma \cap \Gamma').\ Entonces,$$
$$\Gamma'\ \vdash\ \mathrm{M}:\ \sigma\ \text{es derivable}$$

Propiedades de la evaluación - Son teoremas DPPT

Determinismo

$$Si\ M \to N_1\ y\ M \to N_2.\ Entonces,\ N_1\ =\ N_2$$

Preservación de tipos

$$Si\ \vdash M:\sigma\ y\ M \to N.\ Entonces,\ N:\sigma$$

Progreso

$$Si\ \vdash \mathrm{M}:\sigma.\ Entonces,$$
$$o\ bien\ \text{M es un valor}$$
$$o\ bien\ \exists \text{N tq M} \to \text{N}$$

Terminación

$$Si\ \vdash M:\sigma.\ Entonces\ no\ hay\ una\ cadena\ infinita\ de\ pasos\ tal\ que:$$
$$M \to M_1 \to M_2 \to \ldots$$

Corolarios - Canonicidad (SOLO VALEN PARA CALCULO LAMBDA B?)

$$1)\ Si\ \vdash\ \mathrm{M}:\text{bool es derivable.}$$
$$Entonces\ la\ evaluación\ de\ \text{M termina}\ y\ el\ resultado\ es\ \text{T o F}$$
$$2)\ Si\ \vdash \mathrm{M}:\sigma \to \beta\ \text{es derivable.}\ Entonces\ la\ evaluación\ de\ M\ termina$$
$$y\ el\ resultado\ \text{es una abstracción}$$

Extensión de Pares

M, N := ... | <M, N> | π_1(M) | π_2(M)

$\sigma,\ \tau\ :=\ \ldots \mid \sigma \times \tau$

Tipado

$$\frac{\Gamma\ \vdash\ M:\sigma \qquad \Gamma\ \vdash\ N:\tau}{\Gamma \vdash\ \ < M,\ N >:\ \sigma \times \tau}\ t-tupla \qquad \frac{\Gamma \vdash\ M:\ \sigma \times \tau}{\Gamma\ \vdash\ \pi_1(M):\sigma}\ t-\pi_1 \qquad \frac{\Gamma \vdash\ M:\ \sigma \times \tau}{\Gamma\ \vdash\ \pi_2(M):\tau}\ t-\pi_2$$

Semantica: congruencia y computo

V := ...| <V,V>

$$\frac{M\ \to\ M'}{< M,\ N >\ \to\ < M',\ N >}\ E-tupla \qquad \frac{N\ \to\ N'}{< V,\ N >\ \to\ < V,\ N' >}\ E-tupla2$$

$$\frac{M\ \to\ M'}{\pi_1(M)\ \to\ \pi_1(M')}\ E-\pi_1 \qquad \frac{M\ \to\ M'}{\pi_2(M)\ \to\ \pi_2(M')}\ E-\pi_2$$

$$\frac{}{\pi_1 < V_1,\ V_2 >\ \to V_1}\ E-\pi_1 valor \qquad \frac{}{\pi_2 < V_1,\ V_2 >\ \to V_2}\ E-\pi_2 valor$$

Extension Uniones disjuntas

τ ::= . . . | τ + τ

M ::= . . . | $left_\tau$(M) | $right_\tau$(M) | case M of left(x) $\rightsquigarrow$ M [] right(y) $\rightsquigarrow$ M

$$\frac{\Gamma \vdash M : \tau}{\Gamma \vdash left_{\sigma}(M) : \tau + \sigma}\ t - left \qquad \frac{\Gamma \vdash M : \tau}{\Gamma \vdash rigth_{\sigma}(M) : \sigma + \tau} t - rigth$$

$$ej\, \Gamma \vdash left_{Nat \to Bool}(Zero) : Nat + (Nat \to Bool)$$

$$\frac{\Gamma \vdash M : \sigma + \tau \quad \Gamma, x : \sigma \vdash N : \rho \quad \Gamma, y : \tau \vdash O : \rho}{\Gamma \vdash case\, M\, of\, left(x) \rightsquigarrow N \, [] \, right(y) \rightsquigarrow O : \rho} t - case$$

Semántica: Congruencia y computo

V = ...| $left_{\tau}$(M) | $right_{\tau}$(M)|

$$\frac{M \to M'}{left_{\sigma}(M) \to left_{\sigma}(M')}\ E - left \qquad \frac{M \to M'}{right_{\sigma}(M) \to right_{\sigma}(M')}\ E - rigth$$

$$\frac{M \to M'}{case\, M\, of\, left(x) \rightsquigarrow N \, [] \, right(y) \rightsquigarrow O \to case\, M'\, of\, left(x) \rightsquigarrow N \, [] \, right(y) \rightsquigarrow O} E - case$$

$$\frac{}{case\, left_{\sigma}(V)\, of\, left(x) \rightsquigarrow N \, [] \, right(y) \rightsquigarrow O \to N\{x := V\}}\ E - leftCase$$

$$\frac{}{case\, right_{\sigma}(V)\, of\, left(x) \rightsquigarrow N \, [] \, right(y) \rightsquigarrow O \to O\{y := V\}}\ E - rightCase$$

Extension con listas

τ ::= ... | $[\tau]$

$$\begin{aligned} M, N, O ::= \ldots \mid [\ ]_{\tau} \mid M :: N \mid & case\, M\, of\, \{[\ ] \rightsquigarrow N \mid h :: t \rightsquigarrow O\} \\ \mid & foldr\, M\, base\ N ;\ rec(h, r) \rightsquigarrow O \\ \mid & map(M, N) \\ \mid & [M \mid x \leftarrow S,\ P] \end{aligned}$$

$$\frac{}{\Gamma \vdash []_{\tau} : \tau} t - vacio \qquad \frac{\Gamma \vdash M : \sigma \qquad \Gamma \vdash N : [\sigma]}{\Gamma \vdash M :: N : [\sigma]} t - append$$

$$\frac{\Gamma \vdash M : [\sigma] \quad \Gamma \vdash N : \rho \quad \Gamma, h : \sigma, t : [\sigma] \vdash O : \rho}{\Gamma \vdash case\, M\, of\, \{[] \rightsquigarrow N \mid h :: t \rightsquigarrow O\} : \rho} t - case$$

$$\frac{\Gamma \vdash M : [\sigma] \quad \Gamma \vdash N : \rho \quad \Gamma, h : \sigma, r : [\sigma] \vdash O : \rho}{\Gamma \vdash foldr\, M\, base\ N ;\ rec(h, r) \rightsquigarrow O : \rho} t - fold$$

$$\frac{\Gamma \vdash M : \sigma \to \tau \quad \Gamma \vdash N : [\sigma]}{\Gamma \vdash map(M, N) : [\tau]} t - map$$

$$\frac{\Gamma, x : \sigma \vdash M : \rho \qquad \Gamma \vdash S : [\sigma] \qquad \Gamma, x : \sigma \vdash P : Bool}{\Gamma \vdash [M \mid x \leftarrow S,\ P] : [\rho]} t - listasComp$$

Semantica

$V, W \;=\; \ldots \mid []_\sigma \mid V :: W$

$$\frac{M \to M'}{M :: N \to \; M' :: N} E-cabeza \qquad \frac{N \to N'}{V :: N \to \; V :: N'} E-cola$$

$$\frac{M \;\to M'}{case\, M\, of\, \{[] \rightsquigarrow \; N \mid h \;::\; t \rightsquigarrow O\} \quad \to \quad case\, M'\, of\, \{[] \rightsquigarrow \; N \mid h \;::\; t \rightsquigarrow O\}} E-case$$

$$\frac{}{case\, []_\sigma\, of\, \{[] \rightsquigarrow \; N \mid h \;::\; t \rightsquigarrow O\} \;\to\; N} E-vacio$$

$$\frac{}{case\, V1 :: V2\, of\, \{[] \rightsquigarrow \; N \mid h \;::\; t \rightsquigarrow O\} \;\to\; O\{h := V1,\; t := V2\}} E-noVacio$$

$$\frac{M \to M'}{foldr\, M\, base\;\; N\,;\; rec(h,\; r) \;\rightsquigarrow\; O \;\to\; foldr\, M'\, base\;\; N\,;\; rec(h,\; r) \;\rightsquigarrow\; O} E-fold$$

$$\frac{}{foldr\, base\, base\;\; N\,;\; rec(h,\; r) \;\rightsquigarrow\; O \;\to N} E-base$$

$$\frac{}{foldr\; V_1 :: V_2\; base\;\; N\,;\; rec(h,\; r) \;\rightsquigarrow\; O \;\to O\{h := V_1,\; r := \; foldr\; V_2\; base\;\; N; rec(h,\; r) \;\rightsquigarrow\; O\}} E-rec$$

$$\frac{M \to M'}{map(M,\; N) \to \; map(M',\; N)} E-map \qquad \frac{N \to N'}{map(V,\; N) \to \; map(V,\; N'} E-map2$$

$$\frac{\Gamma \vdash \; V : \sigma \to \tau}{map(V,\; []_\sigma) \to []_\tau} E-vacio \qquad \frac{\Gamma \vdash \; W : \sigma \to \tau}{map(W,\; V_1 :: V_2) \to W\; V_1 ::\; map(V_0,\; V_2)} E-noVacio$$

$$\frac{S \to S'}{\left[M \mid\, x \leftarrow S,\; P\right] \to \left[M \mid\, x \leftarrow S',\; P\right]} E-listasComp$$

$$\frac{}{\left[M \mid\, x \leftarrow []_\sigma,\; P\right] \to []_\rho} E-vacio$$

$$\frac{}{\left[M \mid\, x \leftarrow V_1 :: V_2,\; P\right] \to if\; P\{x := V_1\}\; then\; M\{x := V_1\} :: \left[M \mid\, x \leftarrow V_2,\; P\right]\; else\; \left[M \mid x \leftarrow V_2,\; P\right]} E-noVac$$

Extensión para colas bidireccionales

$\sigma \;:=\; \ldots \mid Cola_\sigma$

$M \;:=\; \ldots \mid <>_{\;\sigma} \mid M \bullet M \mid proximo(M) \mid desencolar(M) \mid case\; M\; of\;\; <> \;\; \rightsquigarrow M; c \;\bullet x \;\rightsquigarrow M$

$\qquad \mid recr\; M\; \triangleright \langle\rangle \;\rightsquigarrow\; M;\; r, c \bullet x \rightsquigarrow M$

$$\frac{}{\Gamma \vdash <>_{\;\sigma} :\; Cola_\sigma} t-colaVacia \qquad \frac{\Gamma \vdash M : Cola_\sigma \quad \Gamma \vdash N : \sigma}{\Gamma \vdash M \bullet N :\; Cola_\sigma} t-encolar$$

$$\frac{\Gamma \vdash M:\ Cola_\sigma}{\Gamma \vdash proximo(M):\sigma}t-proximo \qquad \frac{\Gamma \vdash M:Cola_\sigma}{\Gamma \vdash desencolar(M):\ Cola_\sigma}t-desencolar$$

$$\frac{\Gamma \vdash M:\ Cola_\sigma \quad \Gamma \vdash N:\ \rho \quad \Gamma,\ c:Cola_\sigma,\ x:\sigma \vdash O:\rho}{\Gamma \vdash case\ M\ of\ \ <>\ \rightsquigarrow N; c\ \bullet x\ \rightsquigarrow O:\ \rho}t-case$$

$$\frac{\Gamma \vdash M:\ Cola_\sigma \quad \Gamma \vdash N:\ \rho \quad \Gamma, r:\rho,\ c:Cola_\sigma,\ x:\sigma \vdash O:\rho}{\Gamma \vdash recr\ M\ \triangleright \langle\rangle\ \rightsquigarrow\ N;\ r, c \bullet x \rightsquigarrow O:\ \rho}t-recr$$

V:= ... | $<>_\sigma \mid V \bullet V$

encolar, proximo, desencolar y case(Solo para M, N y O no) tienen reglas de congruencia.
Proximo y desencolar solo tienen sentido cuando la cola no es vacia. Entonces, hay que usar
$<>_\sigma\ \bullet V\ ;\ V \bullet V \bullet V$

Reglas de reduccion(computo).

$$\frac{}{proximo(\ <>_\sigma\ \bullet V)\ \to\ V}E-proxBorde \qquad \frac{}{proximo(V_1 \bullet V_2 \bullet V_3)\ \to\ proximo(V_1 \bullet V_2)}E-proximo$$

$$\frac{}{desencolar(\ <>_\sigma\ \bullet V)\ \to\ \ <>_\sigma}E-desencolarBorde$$

$$\frac{}{desencolar(V_1 \bullet V_2 \bullet V_3)\ \to\ desencolar(V_1 \bullet V_2) \bullet V_3}E-desencolar$$

$$\frac{}{case\ \ <>_\sigma of\ \ <>\ \ \rightsquigarrow N; c\ \bullet x\ \rightsquigarrow O\ \to N}E-casoVacio$$

$$\frac{}{case\ V_1 \bullet V_2\ of\ \ <>\ \ \rightsquigarrow N; c\ \bullet x\ \rightsquigarrow O\ \to O\{c := V_1,\ x := V_2\}}E-casoNoVacio$$

$$\frac{}{recr\ \langle\rangle_\sigma\ \triangleright \langle\rangle\ \rightsquigarrow\ N;\ r, c \bullet x \rightsquigarrow O \to N}E-recrBase$$

$$\frac{}{\begin{array}{c}recr\ V \bullet W\ \triangleright \langle\rangle\ \rightsquigarrow\ N;\ r, c \bullet x \rightsquigarrow O \to \\ O\{x := W,\ c := V,\ r := recr\ V \triangleright \langle\rangle\ \rightsquigarrow\ N;\ r, c \bullet x \rightsquigarrow O\}\end{array}}E-recr$$

$$\begin{array}{c}case\ \langle\rangle_{Nat} \bullet\ 1\ \bullet\ 0\ of\ \langle\rangle\ \rightsquigarrow\ próximo(\langle\rangle Bool);\ c\ \bullet\ x\ \rightsquigarrow\ isZero(x)\ \to\ E-case \\ isZero(x)\{x := 0,\ c := \langle\rangle_{Nat} \bullet\ 1\}\ \to\ E-isZeroTrue \\ true\end{array}$$

$$ultimo_\sigma\ =_{def} (\lambda t:Cola_\sigma.\, case\ t\ of\ \ <>_\sigma \rightsquigarrow\ \ \perp_\sigma$$
$$c\ \bullet\ x\ \ \rightsquigarrow\ x)$$

$\vdash ultimo_\sigma:\ Cola_\sigma \to \sigma$ juicio de tipado de la macro

Árboles binarios

$\sigma\ :=\ ... | AB_\sigma$

$M, N, O :=\ ...\quad |Nil_\sigma|\ Bin(M,\ N,\ O)\ |\ raiz(M)\ |\ der(M)|\ izq(M)|\ esNil(M)$
$\quad |\ case\ M\ of\ Nil\ \rightsquigarrow\ N;\ Bin(i,\ r, d)\ \rightsquigarrow\ O$
$\quad |\ map(M, N)\ N\ arbol,\ M\ funcion$

$$\frac{}{\Gamma \vdash Nil_\sigma : AB_\sigma} t-nil \quad \frac{\Gamma \vdash M : AB_\sigma}{\Gamma \vdash raiz(M) : \sigma} t-raiz \quad \frac{\Gamma \vdash M : AB_\sigma}{\Gamma \vdash izq(M) : AB_\sigma} t-izq \quad \frac{\Gamma \vdash M : AB_\sigma}{\Gamma \vdash der(M) : AB_\sigma} t-der$$

$$\frac{\Gamma \vdash M : AB_\sigma}{\Gamma \vdash esNil(M) : Bool} t-esNil \qquad \frac{\Gamma \vdash M : AB_\sigma \qquad \Gamma \vdash N : \sigma \qquad \Gamma \vdash O : AB_\sigma}{\Gamma \vdash Bin(M,\ N,\ O) : AB_\sigma} t-esNil$$

$$\frac{\Gamma \vdash M : AB_\sigma \quad \Gamma \vdash N : \rho \qquad \Gamma,\ i : AB_\sigma, r : \sigma,\ d : AB_\sigma \vdash O : \rho}{\Gamma \vdash case\ M\ of\ Nil\ \rightsquigarrow\ N;\ Bin(i,\ r, d)\ \rightsquigarrow\ O :\ \rho} t-case$$

$$\frac{\Gamma \vdash M : \sigma \to \tau \quad \Gamma \vdash N : AB_\sigma}{\Gamma \vdash map(M,\ N) : AB_\tau} t-map$$

V:= ...| $Nil_\sigma |\ Bin(V, V, V)$

$$\frac{M \to M'}{Bin(M,\ N,\ O)\ \to\ Bin(M',\ N,\ O)} E-bin1 \qquad \frac{M \to M'}{raiz(M) \to\ raiz(M')} E-raiz$$

$$\frac{N \to N'}{Bin(V,\ N,\ O)\ \to\ Bin(V,\ N',\ O)} E-bin2 \qquad \frac{M \to M'}{izq(M) \to\ izq(M')} E-izq$$

$$\frac{O \to O'}{Bin(V_1,\ V_2,\ O)\ \to\ Bin(V_1,\ V_2,\ O')} E-bin3 \qquad \frac{M \to M'}{der(M) \to\ der(M')} E-der$$

$$\frac{M \to M'}{esNil(M) \to\ esNil(M')} EesNil \quad \frac{}{esNil(Nil_\sigma) \to True} EesNilTru \quad \frac{}{esNil(Bin(V_1, V_2, V_3)) \to False} EesNilFalse$$

$$\frac{}{raiz(Bin(V_1, V_2, V_3)) \to\ V_2} E-raizBin \qquad \frac{}{izq(Bin(V_1, V_2, V_3)) \to\ V_1} E-izqBin$$

$$\frac{}{der(Bin(V_1, V_2, V_3)) \to\ V_3} E-derBin$$

$raiz(Nil_\sigma),\ der(Nil_\sigma),\ izq(Nil_\sigma)$ no existen porque perdemos el progreso.

$$\frac{M \to M'}{case\ M\ of\ Nil\ \rightsquigarrow\ N;\ Bin(i,\ r, d)\ \rightsquigarrow\ O \to case\ M'\ of\ Nil\ \rightsquigarrow\ N;\ Bin(i,\ r, d)\ \rightsquigarrow\ O} E-case$$

$$\frac{}{case\ Nil_\sigma\ of\ Nil\ \rightsquigarrow\ N;\ Bin(i,\ r, d)\ \rightsquigarrow\ O \to\ N} E-caseNil$$

$$\frac{}{case\ Bin(V_1,\ V_2,\ V_3)\ of\ Nil\ \rightsquigarrow\ N;\ Bin(i,\ r, d)\ \rightsquigarrow\ O \to\ O\{i := V_1,\ r := V_2,\ d := V_3\}} E-caseBin$$

$$\frac{M \to M'}{map(M,\ N)\ \to\ map(M',\ N)} E-map1 \quad \frac{N \to N'}{map(V,\ N)\ \to\ map(V,\ N')} E-map2$$

$$\frac{\Gamma \vdash V : \sigma \to \tau}{map(V,\ Nil_\sigma)\ \to Nil_\tau} E - mapNil \qquad \frac{\Gamma \vdash V : \sigma \to \tau}{map(V, Bin(I,\ r,\ D))\ \to Bin(map(V,\ I),\ V\,r,\ map(V, D))} E - mapBin$$

$$definimos\ macros$$
$$esNil_\sigma\ :=\ \lambda x : AB_\sigma .\ case\ x\ of\ Nil_\sigma\ \rightsquigarrow True\ ; Bin(i, r, d)\ \rightsquigarrow False$$
$$izq_\sigma\ :=\ \lambda x : AB_\sigma .\ case\ x\ of\ Nil_\sigma\ \rightsquigarrow \perp_{AB\sigma} ; Bin(i, r, d)\ \rightsquigarrow i$$
$$raiz_\sigma\ :=\ \lambda x : AB_\sigma .\ case\ x\ of\ Nil_\sigma\ \rightsquigarrow \perp_{\sigma} ; Bin(i, r, d)\ \rightsquigarrow r$$

Árboles ternarios

$\sigma :=\ ...\ |\ AT(\sigma)$

$M\ :=\ ...\ |\ Tnil_\sigma\ |\ Tern(M, M, M, M)\ |\ foldAT\ M\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O$

$$\frac{}{\Gamma \vdash Tnil_\sigma : AT(\sigma)} t - Nil \qquad \frac{\Gamma \vdash M_1 : \sigma \quad \Gamma \vdash M_2 : AT(\sigma) \quad \Gamma \vdash M_3 : AT(\sigma) \quad \Gamma \vdash M_4 : AT(\sigma)}{\Gamma \vdash Tern(M_1, M_2, M_3, M_4) : AT(\sigma)} t - tern$$

$$\frac{\Gamma \vdash M : AT(\sigma) \quad \Gamma \vdash N : \rho \quad \Gamma, x : \sigma, ri : AT(\sigma), rm : AT(\sigma), rd : AT(\sigma) \vdash O : \rho}{foldAT\ M\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O : \rho} t - fold$$

$V := ...\ |\ Tnil_\sigma\ |\ Tern(V, V, V, V)$

Reduccion

Congruencia→ para los 4 terminos de tern de izquierda a derecha

fold el del medio M

Computo

$$\frac{}{foldAT\ Tnil_\sigma\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O \to\ N} t - foldNil$$

$$\frac{}{\begin{array}{c} foldAT\ Tern(V_1,\ V_i,\ V_m, V_d)\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O \to \\ O\{x := V1,\ ri :\ foldAT\ V_i\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O, \\ rm :\ foldAT\ V_m\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O, \\ rd :\ foldAT\ V_d\ \rhd\ Tnil \rightsquigarrow N;\ Tern(x,\ ri,\ rm, rd)\ \rightsquigarrow O\} \end{array}} t - foldTern$$